

3 Tire Architecture (Web>App>DB)

- ✓ Created **Web01** and installed (**Nginx**) into it and **Nsg** (source:**Any**, Dest: **App01**, DPort:**80**) and (SSH:**22**)
- ✓ Next **App01** installed (**Tomcat**)
- ✓ Allow port from Source: **Web01**(private Ip), Dest:**APP01**(Private Ip), Dport:**8080**

Home > Compute infrastructure | Virtual machines > Uapp01

Uapp01 | Network settings

Virtual machine

What are the requirements for attaching and detaching network interfaces? List all my network interfaces for this VM +1

Search

Network security group **Uapp01-nsg** (attached to networkInterface: **uapp01401**)
Impacts 0 subnets, 1 network interfaces

+ Create port rule

Search rules

Source == all Destination == all Protocol == all Action == all Port == all

Prio...	Name	Port	Protocol	Source	Destination	Action
Inbound port rules (5)						
310	allow-app	8080	Any	Any	172.16.0.4	Allow
320	ssh-allow	22	Any	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInB...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound port rules (3)						

13:14 31/01/2026

TOMCAT

To install **Apache Tomcat** on **Ubuntu**, follow these steps:

To switch from a **normal user** (\$ prompt) to the **root user** (# prompt) in Ubuntu, use any of the following commands:

Method 1: Using `sudo su` (Recommended)

- `sudo su`
 - a. This switches you to the **root user**.
 - b. The prompt will change from \$ to #.

Step 1: Update System Packages

- `apt update`

Step 2: Install Java

Tomcat requires Java. Install OpenJDK:

- `apt install default-jdk -y`

Verify the installation:

- `java -version`

Step 3: Download Tomcat

- Go to the [Apache Tomcat official website](https://tomcat.apache.org/) and get the latest version.

Run:

- `wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.18/bin/apache-tomcat-10.1.18.tar.gz`

(Replace the version number with the latest one if needed.)

- `ls` (to list the files)

Step 4: Extract and Configure Tomcat

- `tar -xvzf apache-tomcat-10.1.18.tar.gz`
- `ls`
- `mv apache-tomcat-10.1.18 tomcat`
- `ls`

To start Apache Tomcat manually from the **bin** directory, follow these steps:

Step 1: Navigate to the Tomcat Directory

- `cd /tomcat/bin`

Step 2: Start Tomcat

Run the startup script: (`/` is to execute a command)

- `./startup.sh`

Step 4: Access Tomcat

Open your browser and go to:

http://your_server_ip:8080

➤ Editing context.xml for Host Manager:

The Valve element in Tomcat's context.xml file is used for **access control and request filtering**. Specifically, the RemoteAddrValve restricts access based on the client's IP address.

Use of Valve in context.xml (META-INF)

Purpose:

- Controls access to web applications (like manager and host-manager).
- Allows or denies requests from specific IP addresses.
- Enhances security by limiting administrative access.

104.211.101.190 (azureuser)

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...

/home/azureuser/

Name

..

.cache

.ssh

.bash_logout

.bashrc

.profile

.Xauthority

Remote monitoring

Follow terminal folder

```
update-alternatives: using /usr/lib/jvm/java-21-openjdk-amd64/bin/serialver to provide /usr/bin/serialver (serialver) in auto mode
update-alternatives: using /usr/lib/jvm/java-21-openjdk-amd64/bin/jhsdb to provide /usr/bin/jhsdb (jhsdb) in auto mode
Setting up default-jre-headless (2:1.21-75+exp1) ...
Setting up default-jre (2:1.21-75+exp1) ...
Setting up openjdk-21-jdk:amd64 (21.0.9+10-1~24.04) ...
update-alternatives: using /usr/lib/jvm/java-21-openjdk-amd64/bin/jconsole to provide /usr/bin/jconsole (jconsole) in auto mode
Setting up default-jdk-headless (2:1.21-75+exp1) ...
Setting up default-jdk (2:1.21-75+exp1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.42.10+dfsg-3ubuntu3.2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@Uapp01:/home/azureuser# java -version
openjdk version "21.0.9" 2025-10-21
OpenJDK Runtime Environment (build 21.0.9+10-Ubuntu-124.04)
OpenJDK 64-Bit Server VM (build 21.0.9+10-Ubuntu-124.04, mixed mode, sharing)
root@Uapp01:/home/azureuser# https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.18/bin/apache-tomcat-11.0.18.tar.gz
bash: https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.18/bin/apache-tomcat-11.0.18.tar.gz: No such file or directory
root@Uapp01:/home/azureuser#
```

Uapp01 0% 0.55 GB / 7.76 GB 0.01 Mb/s 0.00 Mb/s 13 min azureuser (x2) /: 9% /boot: 8% /boot/efi: 6% /mnt: 1%

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104.211.101.190 (azureuser)

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Quick connect...

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Remote monitoring

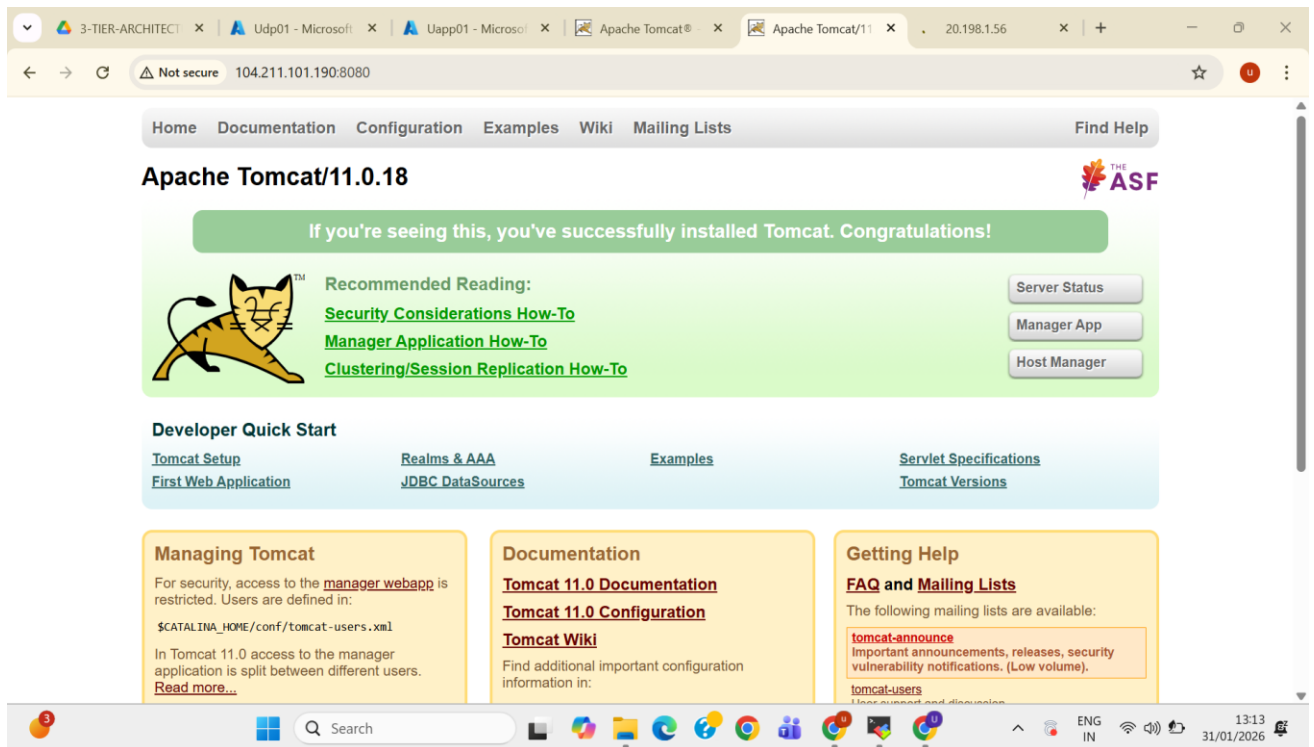
Follow terminal folder

```
apache-tomcat-11.0.18/bin/configtest.sh
apache-tomcat-11.0.18/bin/daemon.sh
apache-tomcat-11.0.18/bin/digest.sh
apache-tomcat-11.0.18/bin/makebase.sh
apache-tomcat-11.0.18/bin/migrate.sh
apache-tomcat-11.0.18/bin/setclasspath.sh
apache-tomcat-11.0.18/bin/shutdown.sh
apache-tomcat-11.0.18/bin/startup.sh
apache-tomcat-11.0.18/bin/tool-wrapper.sh
apache-tomcat-11.0.18/bin/version.sh
root@Uapp01:~# ls
apache-tomcat-11.0.18  apache-tomcat-11.0.18.tar.gz
root@Uapp01:~# mv apache-tomcat-11.0.18 tomcat
root@Uapp01:~# ls
apache-tomcat-11.0.18.tar.gz  tomcat
root@Uapp01:~# cd /tomcat/bin
bash: cd: /tomcat/bin: No such file or directory
root@Uapp01:~# ls
apache-tomcat-11.0.18.tar.gz  tomcat
root@Uapp01:~# cd tomcat/
root@Uapp01:~/tomcat# ls
BUILDING.txt  LICENSE  README.md  RUNNING.txt  conf  logs  webapps
CONTRIBUTING.md  NOTICE  RELEASE-NOTES  bin  lib  temp  work
root@Uapp01:~/tomcat# cd bin/
root@Uapp01:~/tomcat/bin# ls
bootstrap.jar  ciphers.sh  commons-daemon-native.tar.gz  daemon.sh  digest.sh  migrate.bat  shutdown.sh  tool-wrapper.bat
catalina.bat  commons-daemon.jar  configtest.bat  makebase.bat  setclasspath.bat  startup.sh  tomcat-juli.jar  tool-wrapper.sh
catalina.sh  configtest.sh  makebase.sh  shutdown.bat  tomcat-native.tar.gz  version.bat  version.sh
root@Uapp01:~/tomcat/bin#
```

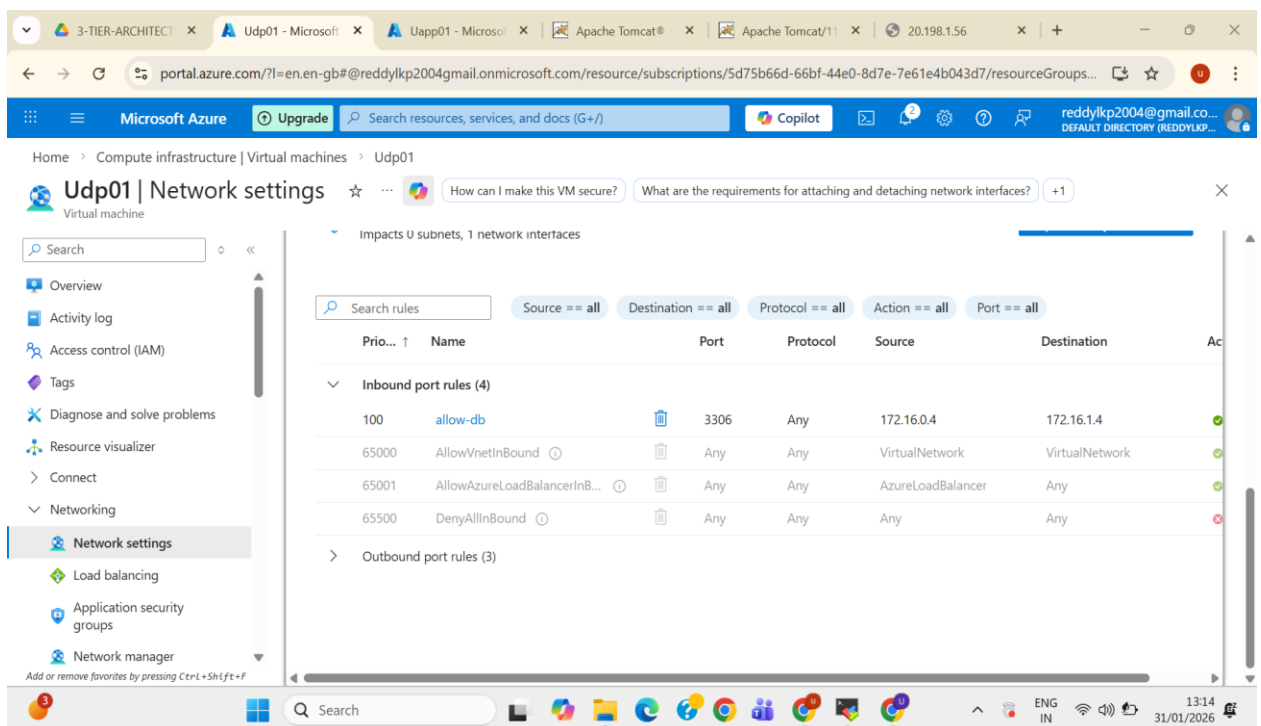
Uapp01 1% 0.57 GB / 7.76 GB 0.10 Mb/s 0.03 Mb/s 31 min azureuser (x2) /: 10% /boot: 8% /boot/efi: 6% /mnt: 1%

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- ✓ Next **DB01** Install (**Mysqldb**)
- ✓ Allow port from Source: **App01**(private Ip),
Dest: **DB01**(Private Ip), Dport: **3306**



To install MYSQL on **Ubuntu**, follow these steps:

To switch from a **normal user** (\$ prompt) to the **root user** (# prompt) in Ubuntu, use any of the following commands:

Method 1: Using sudo su (Recommended)

- `sudo su`
 - c. This switches you to the **root user**.
 - d. The prompt will change from \$ to #.

Step 1: Update System Packages

- `apt update`
 - Run the following command to install MySQL:
- `apt install mysql-server -y`
- To improve security, run the following command:
- `mysql_secure_installation`
- Edit the MySQL configuration file using a text editor:
- `nano /etc/mysql/mysql.conf.d/mysqld.cnf`
- Locate the following line: `bind-address = 127.0.0.1`
- `bind-address = 0.0.0.0`
- EX: Modify it based on your requirement:

Scenario	Configuration
Listen on all interfaces (for remote access)	<code>bind-address = 0.0.0.0</code>
Listen on a specific IP (e.g., 192.168.1.100)	<code>bind-address = 192.168.1.100</code>

- Save and exit (CTRL + X, then Y, then Enter).
- Apply the changes by restarting MySQL:
- `systemctl restart mysql`
- Check if MySQL is running:
- `systemctl status mysql`

After installing mysql on DB, Edit the file which is stored inside `/etc/mysql/mysql.conf.d/mysqld.cnf` ---This is the file we have to edit, In side file there is a **bind-address** it should be **0.0.0.0**,

bind-address = 0.0.0.0 and save & Exit.

- ✓ Now see **connection** from **App01** to **DB01** with the help of **Telnet DB (private ip) (3306)** from **App0**, Because DB we can't browse on internet, That's way we should connect from App to DB. **Status** should be **Connected**.
- ✓ Next configure **Denny** access from **Web01** to **DB01**,
- ✓ No-direct Connection b/w them

20.198.1.56 (azureuser)

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Quick connect...

/home/azureuser/

Name

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Remote monitoring

Follow terminal folder

2. 104.211.101.190 (azureuser)

3. 20.198.1.56 (azureuser)

Memory: 384.4M (peak: 386.4M)
CPU: 13.673s
CGroup: /system.slice/mysql.service
└─18061 /usr/sbin/mysqld

Jan 31 06:33:44 Udp01 systemd[1]: Starting mysql.service - MySQL Community Server...

Jan 31 06:33:46 Udp01 systemd[1]: Started mysql.service - MySQL Community Server.

root@Udp01:/etc/mysql/mysql.conf.d# cd ..

root@Udp01:/etc/mysql# cd ..

root@Udp01:/etc# cd ..

root@Udp01:/# ls

bin boot etc lib lib64 media opt root sbin snap sys usr

bin.usr-is-merged dev home lib.usr-is-merged lost+found mnt proc run sbin.usr-is-merged srv tmp var

root@Udp01:/# cd home/

root@Udp01:/home# ls

LS: command not found

root@Udp01:/home# ls

azureuser

root@Udp01:/home# cd azureuser/

root@Udp01:/home/azureuser# ls

root@Udp01:/home/azureuser# telnet 172.16.1.4 3306

Trying 172.16.1.4...

Connected to 172.16.1.4.

Escape character is '^]'.

UHost 'udp01.internal.cloudapp.net' is not allowed to connect to this MySQL serverConnection closed by foreign host.

root@Udp01:/home/azureuser# telnet 172.16.0.4 8080

Trying 172.16.0.4...

Connected to 172.16.0.4.

Escape character is '^]'.

Connection closed by foreign host.

root@Udp01:/home/azureuser#

Udp01 0% 0.91 GB / 7.75 GB 0.02 Mb/s 0.01 Mb/s 118 min azureuser (x2) /: 9% /boot: 8% /boot/efi: 6% /mnt: 0%

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104.211.101.190 (azureuser)

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2. 104.211.101.190 (azureuser)

3. 20.198.1.56 (azureuser)

Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1017-azure x86_64)

* Documentation: <https://help.ubuntu.com>
* Management: <https://landscape.canonical.com>
* Support: <https://ubuntu.com/pro>

System information as of Sat Jan 31 07:34:50 UTC 2026

System load: 0.0 Processes: 138
Usage of /: 9.1% of 28.02GB Users logged in: 0
Memory usage: 6% IPv4 address for eth0: 172.16.0.4
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: sudo pro status

Last login: Sat Jan 31 05:01:06 2026 from 49.206.40.158

azureuser@Uapp01:~\$ sudo su

root@Uapp01:/home/azureuser# telnet 172.16.1.4 3306

Trying 172.16.1.4...

Connected to 172.16.1.4.

Escape character is '^]'.

UHost 'uapp01.internal.cloudapp.net' is not allowed to connect to this MySQL serverConnection closed by foreign host.

root@Uapp01:/home/azureuser#

Uapp01 0% 0.73 GB / 7.76 GB 0.01 Mb/s 0.00 Mb/s 167 min azureuser (x2) /: 10% /boot: 8% /boot/efi: 6% /mnt: 0%

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